

a_09_implementation_history_and_project_s

SUMMARY

This document records the implementation history of WRA. Unlike the previous documents that describe the future platform, this document describes what has actually been built, what currently exists, what is partially implemented, and what remains unfinished.

PROJECTS

WeRepairRefrigerators | Create a better CRM for appliance repair technicians. The project evolved into a platform: Customer Marketplace + Technician Operating System + AI Advisor + Community + Vendor Search + Inventory Platform.

- M1 Authentication — COMPLETE: Authentication, User Accounts, Role-Based Access, Protected Dashboard Routes, Technician Access, Admin Access
- M2 Technician Profiles — COMPLETE: Technician Profile, Business Name, Experience, ZIP Coverage, Specialties, Public Profile Support
- M3 Public Technician Pages — COMPLETE: Public Technician Profiles, Technician Directory, Profile Slugs, Marketplace Visibility
- M4 Service Requests — COMPLETE: Job Creation, Service Request Storage, Lead Tracking, Status Updates
- M5 Notes System — COMPLETE: Notes, Timeline Integration, Diagnostic Notes, Internal Notes
- M6 Photos — COMPLETE: Photo Uploads, Photo History, Job Attachments
- M7 Estimates — COMPLETE: Estimate Builder, Estimate Editing, Estimate Sending, Estimate Storage
- M8 Customer Approval Flow — COMPLETE: Approval Page, Approval Tokens, Customer Approval, Customer Decline, Automatic Status Updates
- M9 Invoice Foundation — PARTIAL: Invoice Creation, Invoice Storage, Invoice Display; missing Payment Engine, Receipt Engine
- M10 Scheduling Foundation — COMPLETE: Scheduling Engine, Availability Rules, Time Windows, Dispatcher Logic
- M11 Dispatcher Preview — COMPLETE: Technician Matching, Dispatcher Preview, Recommendation Logic, Eligibility Rules
- M12 Technician Availability — COMPLETE: Availability Rules, Working Hours, Availability Storage
- M13 Appointment Booking — COMPLETE: Appointment Creation, Conflict Prevention, Booking Logic
- M14 Calendar Foundation — PARTIAL: Internal Calendar, Google Calendar Sync Foundation, not fully validated
- M15 Job Lifecycle — COMPLETE: New, Contacted, Scheduled, Diagnosed, Estimate Sent, Estimate Approved, Parts Needed, Parts Ordered, Parts Received, Return Visit Scheduled, Completed, Closed, Waiting Customer, Canceled
- M16 Dashboard Redesign — COMPLETE: Large UI effort
- Recent tasks 130–141: Google Calendar Foundation, Dispatch Board, Job Status Lifecycle, Dashboard Style Transition, Job Workspace Redesign, Jobs Inbox Redesign, Dashboard Alignment, Dashboard Data Validation, Dashboard Expansion, Dashboard Density Improvements, Dashboard Compact Layout, Premium Dashboard Pass
- Current dashboard status: 80–90% complete; future work should focus on functionality rather than cosmetic redesigns
- Current priority: Task 143 — End-to-End Workflow Audit + Customer Foundation

TECHNICAL SKILLS

Next.js

TypeScript

Supabase

PostgreSQL

REPOSITORY INFORMATION

Primary Project Location: /Users/serhiitarenko/Desktop/WeRepairRefrigerators

CURRENT CORE DIRECTORIES

Frontend: frontend/; Backend: backend/; Database: supabase/; Documentation: docs/; Infrastructure: infrastructure/

CRITICAL FILES

frontend/src/app/dashboard/page.tsx; frontend/src/components/dashboard/DashboardJobsOverview.tsx; frontend/src/components/dashboard/ServiceRequestsInbox.tsx; frontend/src/components/dashboard/ServiceRequestDetail.tsx; frontend/src/components/dashboard/TechnicianSchedule.tsx; frontend/src/components/dashboard/DashboardSidebar.tsx; frontend/src/components/dashboard/DashboardNavigationLinks.tsx; frontend/src/lib/supabase/types.ts; frontend/src/lib/service-request-records.ts; frontend/src/lib/address-autocomplete.ts

CURRENT TECHNICAL REALITY

Implemented and Usable: Authentication, Technician Profiles, Public Profiles, Jobs, Notes, Photos, Estimates, Estimate Approval, Scheduling, Appointments, Status Lifecycle, Dashboard, Job Workspace, Jobs Inbox. Partially Implemented: Invoices, Calendar Sync, Dispatcher, Dashboard Widgets. Not Implemented: Customer Database, Customer Portal, Appliance Registry, Inventory Backend, Vendor APIs, Community Backend, AI Backend, Payment Processing, Receipt Delivery, Technician Messaging, Voice Rooms, Truck Inventory.

ARCHITECTURAL DECISIONS

Dashboard = Technician Command Center (not analytics dashboard). Scheduling: CRM Appointment Table = source of truth; Google Calendar = outbound sync only. Customer is a first-class entity. Mobile first, desktop secondary. Community is core platform functionality. AI Advisor is a core platform layer.

WORKFLOW AUDIT SCOPE

Customer → Service Request → Appointment → Estimate → Approval → Invoice → Payment → Receipt → Closed Job. Every step must be verified.

KNOWN TECHNICAL GAPS

Customer Database, Customer Portal, Appliance Registry, Payment Engine, Receipt Engine, Vendor Search Backend, Inventory Backend, Community Backend, AI Backend, Marketplace Matching Engine.

ROLLBACK STRATEGY

Before large changes: create milestone tag, document rollback point, update PROJECT_STATE.md, ROADMAP.md, DEVELOPER_HANDOFF.md.

CURRENT DEVELOPMENT PHILOSOPHY

Stop building isolated features. Focus on complete workflows. Focus on mobile-first execution. Focus on customer-to-payment lifecycle. Focus on real-world usability.

SUCCESS DEFINITION

The platform is successful when a customer can create an account, request service, approve an estimate, pay an invoice, and review repair history; and a technician can receive a job, diagnose, estimate, repair, invoice, receive payment, and close the job entirely from a mobile device.

RELATIONSHIP TO FINAL DOCUMENT

Document 10: Execution Roadmap and Future Development Plan